

NLP Assignment 2

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Abstract

This paper presents a cascaded speech-to-text translation system for the low-resource Irish-to-English language pair, developed as part of the Natural Language Processing module for the M.Sc. in AI at MTU. It addresses the challenges of translating Irish (Gaeilge) speech while implementing a two-stage pipeline combining Automatic Speech Recognition (ASR) and Machine Translation (MT) components. The baseline system integrates OpenAI’s Whisper model (Radford et al., 2023) for ASR with Meta’s NLLB-200 (NLLB Team et al., 2022) for neural machine translation. To address the low-resource nature of Irish, we develop an improved system utilising Irish-specific fine-tuned Whisper models that better handle the language’s unique phonological characteristics. We evaluate both systems on the IWSLT 2026 Irish-English test set using standard metrics including BLEU (Papineni et al., 2002) and chrF++ (Popović, 2015), with comprehensive error analysis identifying common failure modes such as language misidentification and morphological errors. Our results demonstrate the effectiveness of language-specific model adaptation in low-resource settings, while highlighting persistent challenges in cascaded architectures including error propagation from ASR to MT stages.

1 Problem Statement

This work addresses the challenge of low-resource speech-to-text translation for the Irish-to-English language pair. Speech translation (ST) — converting spoken audio in one language into text in another — presents significant difficulties for low-resource languages like Irish (Gaeilge), which has limited transcribed audio data and fewer high-quality pretrained models compared to well-resourced languages.

We implement a cascaded pipeline architecture combining Automatic Speech Recognition

(ASR) and Machine Translation (MT) components. This approach offers practical advantages in low-resource settings: ASR and MT components can be independently pretrained on large datasets, errors can be diagnosed at each stage, and strong off-the-shelf models are readily available. Our baseline system integrates OpenAI’s Whisper model for ASR with Meta’s NLLB-200 for neural machine translation.

The primary research question is: *How can we improve translation quality for low-resource Irish speech by optimizing component selection in a cascaded pipeline?* We evaluate two key dimensions: (1) the impact of Irish-specific fine-tuned ASR models versus general multilingual models, and (2) the comparative performance of multilingual (NLLB-200) versus bilingual (OPUS-MT) translation models. Our evaluation uses the IWSLT 2026 Irish-English dataset (Uppaal et al., 2026), measuring performance through BLEU, chrF++, and COMET metrics while conducting systematic error analysis to identify failure modes specific to Irish morphology and phonology.

2 Dataset

We utilize the IWSLT 2026 Irish-English speech translation dataset (Uppaal et al., 2026), which provides parallel Irish speech and English text for low-resource speech translation research. For this work, we exclusively use the development split located in `iwslt2026_ga-eng/iwslt2025_ga-eng/dev/`, as it is the only subset providing both Irish audio files (`wav/`) and corresponding English reference translations (`txt/`).

2.1 Dataset Composition

The development set contains 1120 audio samples with corresponding English translations. This limited dataset size reflects the low-resource nature of Irish speech translation and necessitates reliance on

pretrained models rather than training from scratch. While the dataset includes additional splits (training and test sets), these were not utilised in our pipelines due to the absence of paired English translations required for evaluation.

2.2 Audio Characteristics

All audio files are provided in WAV format with 16kHz sampling rate, mono channel configuration. Audio segments vary in duration, with filenames set so that an alphabetical listing matches the order of the translations in the dev.eng. The dataset includes both individual utterances and slightly longer continuous speech segments, reflecting realistic single sentence speech translation scenarios, but not longer conversations.

2.3 Annotation Format

Ground truth annotations are provided in TSV format (stamped.tsv) containing three fields: audio filename, start timestamp, and length in seconds. In some of the other unused data folders, english reference texts are also available as plain text files (train.eng, dev.eng, test.eng), but in the dev set we only have the dev.eng file for translations.

2.4 Preprocessing

Audio preprocessing is minimal to preserve authentic speech characteristics. We load audio files using the librosa library with 16kHz sampling rate to match Whisper model requirements. No additional noise reduction, normalization, or voice activity detection is applied, as pretrained models are designed to handle raw speech input. For translation evaluation, we tokenize reference texts using standard whitespace splitting to compute BLEU and chrF++ metrics.

2.5 Dataset Challenges

The dataset presents several challenges characteristic of low-resource speech translation: (1) extremely limited training data (1120 samples) necessitating reliance on pretrained models, (2) Irish phonological complexity including lenition, eclipsis, and dialectal variation, and (3) morphological richness with extensive inflection and mutation systems.

3 Pipeline Architectures

We implement four cascaded speech translation systems combining different ASR and MT components. All systems follow a two-stage architecture:

audio → Irish transcription (ASR) → English translation (MT).

3.1 Baseline System: Whisper-Large + NLLB

Our baseline combines OpenAI’s Whisper-large-v3 (Radford et al., 2023) for ASR with Meta’s NLLB-200-distilled-600M (NLLB Team et al., 2022) for MT. Whisper-large-v3 (1.5B parameters) is a multilingual ASR model trained on 680,000 hours of audio across 99 languages including Irish. We use the model without language-specific fine-tuning, relying on its general multilingual capabilities.

For translation, NLLB-200 (600M parameters) supports 200 languages including Irish (gle_Latn) and English (eng_Latn). We configure the model with forced BOS token for target language specification and beam search (num_beams=5) for improved output quality.

Pipeline flow: Audio (16kHz WAV) → Whisper processor (mel-spectrogram extraction, 80 mel bins) → Whisper encoder-decoder → Irish transcript → NLLB tokenizer → NLLB encoder-decoder → English translation.

Justification: This baseline represents a strong starting point using state-of-the-art multilingual models with proven performance across diverse languages. Both models are readily available via HuggingFace and require no additional training.

3.2 Improved System 1: Specialized Whisper + NLLB

The first improvement replaces the general Whisper model with InoWouw/whisper-small-irish, a Whisper-small variant (244M parameters) fine-tuned specifically on Irish speech data. This model addresses the baseline’s tendency to misidentify Irish as Welsh or English due to phonological similarities.

Key modifications:

- Irish-specific acoustic modeling trained on native speaker data
- Reduced model size (244M vs 1.5B parameters) with specialized performance
- Same mel-spectrogram preprocessing (80 bins, 16kHz) for compatibility
- Identical NLLB-200 MT component as baseline

Justification: Language-specific fine-tuning should improve ASR accuracy for Irish phonological features (lenition, eclipsis, broad/slender consonants), reducing error propagation to the MT stage. The smaller model size also improves inference speed.

3.3 Improved System 2: Whisper-Large + OPUS-MT

This variant maintains the baseline Whisper-large-v3 ASR but replaces NLLB with Helsinki-NLP’s opus-mt-ga-en, a bilingual Irish-English translation model from the OPUS-MT family. Unlike NLLB’s multilingual approach, OPUS-MT is trained exclusively on Irish-English parallel data.

Key modifications:

- Bilingual translation model (smaller, 300M parameters)
- Direct Irish→English optimization without multilingual interference
- Simpler tokenization without language-specific BOS tokens
- Same beam search configuration (num_beams=5)

Justification: Bilingual models often outperform multilingual models on specific language pairs due to focused training. OPUS-MT may better handle Irish-specific translation challenges like VSO word order and prepositional pronouns.

3.4 Improved System 3: Specialized Whisper + OPUS-MT

Our best-performing system combines both improvements: Irish-specific ASR (InoWouw/whisper-small-irish) with bilingual MT (opus-mt-ga-en). This architecture optimizes both pipeline stages for the Irish-English language pair.

Justification: Combining specialized ASR and focused MT should provide synergistic improvements. Better Irish transcriptions from the specialized ASR reduce MT errors, while the bilingual MT model better handles the improved Irish input. This represents the most language-pair-specific configuration tested.

3.5 Implementation Details

All systems share common preprocessing: audio resampling to 16kHz mono using librosa, Whisper

feature extraction with 80 mel-frequency bins, and beam search decoding. We process the complete IWSLT 2026 development set (1120 samples) for each system. Models run on GPU (CUDA) with float32 precision to ensure numerical stability. No audio augmentation or text post-processing is applied to maintain fair comparison across systems.

4 Experiment

4.1 Evaluation Setup

Footnotes are inserted with the \footnote command.¹

4.2 Metrics

metrics

4.3 Experimental Conditions

4.4 Citations

You can use the command \citep (cite in parentheses) to get “(author, year)” citations ([Gusfield, 1997](#)).

([Gusfield, 1997](#))

5 BLEU/chrF++ Scores

6 Qualitative Analysis

7 Result Interpretations

8 Summary

References

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Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318.

Maja Popović. 2015. chrF: character n-gram F-score for automatic MT evaluation. In *Proceedings of the Tenth Workshop on Statistical Machine Translation*, pages 392–395.

¹This is a footnote.

Table 1: Overall Comparison Table

System	Samples	Valid	BLEU	chrF++	COMET	Coverage	Rep %	Avg Hyp Len	Avg Ref Len	Length Ratio	Avg Time/File (s)
Specialised ASR+NLLB	1120	1120	4.36	22.25	0.4752	100.0	2.59	10.40	9.88	1.05	0.82
Specialised ASR+OPUS-MT	1120	1120	3.40	21.56	0.4384	100.0	0.62	10.55	9.88	1.07	0.64
Baseline ASR+NLLB	1120	1120	1.51	15.90	0.4375	100.0	4.11	10.31	9.88	1.04	1.70
Baseline ASR+OPUS-MT	1120	1120	1.42	15.19	0.4168	100.0	2.32	10.07	9.88	1.02	1.56

Table 2: Relative Comparison vs Baseline ASR + NLLB

System	BLEU	BLEU Δ vs Baseline	chrF++	chrF++ Δ vs Baseline	Coverage	Coverage Δ vs Baseline	Avg Time/File (s)	Time/File Δ vs Baseline
Specialised ASR + NLLB	4.36	2.85	22.25	6.35	100.0	0.0	0.82	-0.88
Specialised ASR + OPUS-MT	3.40	1.89	21.56	5.66	100.0	0.0	0.64	-1.06
Baseline ASR + NLLB	1.51	0.00	15.90	0.00	100.0	0.0	1.70	0.00
Baseline ASR + OPUS-MT	1.42	-0.09	15.19	-0.71	100.0	0.0	1.56	-0.14

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. *arXiv preprint arXiv:2212.04356*.

Shashwat Uppaal and 1 others. 2026. IWSLT 2026 irish-english speech translation dataset. https://github.com/shashwatup9k/iwslt2026_ga-eng. Accessed: 2026-05-12.

A AI Use Declaration

I used AI to blah blah blah.